## Letter of Amendment #2 dated 04 December 2023

**Study Title:** Innovative machine learning and multi-source data analysis towards development of an urban heat-health Early Warning System for African cities
**Protocol number: HEAT002**
**Date of the last version of the protocol: 15 September 2023**
**HREC Reference No: 200606**

**The overall rationale for the amendment**
The amendments to the ethical considerations of the protocol stem from a meeting with Professor Paul Ruff, chair of the ethics committee, where insightful discussions and recommendations were provided.

**Implementation of the protocol modification**

Upon receiving the research ethics committee's approval for this protocol amendment, the site would implement the protocol amendment immediately. A copy of the protocol amendment approval and any correspondence will be retained in the site's regulatory files of this study.

**All proposed changes and their rationale**

We have edited the text in the following sections of the protocol, described in the table below:

| Location of Revision | Description of Revision | Reason for changes |
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| **6. Ethical and Privacy considerations** | Below is a summary of the changes made to Section 6 of the protocol<br><br>1. We added the following guidelines to the list of legal and ethical guidelines that we adhere to in the study:<br>- US Department of Health and Human Services, Protection of Human Subjects in Research, 45 CFR 46<br>2. We removed any text referencing reconsenting or applying for individual informed consent waivers. This was contained in Table 1, under ethical considerations and the section titled ''Assessment of the informed consent process in the primary study''. We added the following text: "No reconsenting of study | The amendments to Section 6 of the protocol stem from a meeting with Professor Paul Ruff, chair of the ethics committee, where insightful discussions and recommendations were provided.<br><br>The overarching goal is to align the HE$^2$AT Center's research practices with the ethical standards and best practices discussed during this meeting.<br><br>The decision to omit references to reconsenting or individual informed consent waivers reflects the guidance provided during the meeting, with particular reference to the Department of Health Ethics in Health Research Guidelines of 2015. Professor Ruff indicated that pursuing such processes for secondary anonymised data would not be required. |

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| | participants will be conducted for use of secondary anonymised data in the HE$^2$AT Center, which meets the requirements of the ethical guidelines, listed above, that are being adhered to."<br>3. We added clarifying text related to the data-sharing agreement. We noted that data providers would acknowledge that they have addressed their local ethical and legal requirements before sharing the data with the HE$^2$AT Center. Additionally, the data-sharing agreements will be shared with Wits HREC. Lastly, we added that Wits RHI and UCT will assist with finalising data-sharing agreements.<br>4. Lastly, amendments were made to Table 2 to address the approach to the different data sources. We added that Wits HREC would approve studies that had been previously approved by the Wits HREC for use in the HE$^2$AT Center, following a review of the study synopsis, informed consent form, ethics approval letter, and data sharing agreement, where available. We further clarified that data providers outside of Wits are responsible for ensuring they have the required, local ethical, and legal clearances to share the data, as agreed to within the data sharing agreement. | The clarification regarding data sharing agreements is to better ensure that the different ethical and legal conditions from locations outside of Wits University are adhered to, as assured by the data provider and their institution.<br><br>Changes made to Table 2 are nuanced to reflect Professor Ruff's specific guidance during the meeting. The emphasis on external data providers assuring their local clearances aligns with the meeting discussions. Further, this section was amended to align with the approval requirement from Wits HREC for studies approved by the same HREC, following guidance from Prof Ruff.<br><br>Collectively, these amendments underscore a commitment to integrating specific recommendations from the Wits Human Research Ethics Committee into the protocol, ensuring a robust ethical framework for HE$^2$AT Center. |